

RATNAPURA, Sri Lanka

WMO: 434860

Lat: 6.7129N

Lon: 80.3808E

Elev: 88

StdP: 100.27

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	21.0	21.9	18.0	13.0	27.3	19.3	14.2	27.1	5.2	26.9	4.3	27.6	0.2	N/A	0.265

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	6.6	34.1	25.6	33.3	25.7	32.7	25.6	27.4	31.6	27.1	31.1	26.7	30.6	1.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	22.0	29.2	26.0	21.6	28.9	25.7	21.2	28.5	87.4	31.8	86.0	31.2	84.6	30.8	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.5	3.6	3.1	DB	17.0	36.3	2.6	1.7	15.1	37.5	13.5	38.5	12.1	39.4	10.2	40.6	(4)
(5)			WB	16.9	28.2	2.4	0.4	15.2	28.5	13.8	28.7	12.4	28.9	10.6	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	27.0	27.6	28.2	28.2	27.8	27.2	27.3	26.9	26.9	26.9	26.9	26.6	(6)	
	DBStd	1.19	1.26	1.01	0.98	1.05	1.13	1.03	1.04	0.96	1.06	0.99	0.92	1.22	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6308	526	493	565	546	553	516	535	524	506	526	506	513	(10)	
	CDD18.3	3267	268	260	307	296	294	266	277	266	256	267	256	255	(11)	
	CDH23.3	27991	2191	2254	2792	2627	2874	2493	2633	2327	2108	2017	1804	1871	(12)	
(13)	CDH26.7	9217	846	927	1186	1028	951	654	718	589	570	596	557	596	(13)	
(14) Wind	WSAvg	0.8	0.6	0.6	0.7	0.7	0.8	1.3	1.4	1.2	0.9	0.7	0.5	0.5	(14)	
(15) Precipitation	PrecAvg	3716	105	128	205	348	495	390	308	297	421	439	356	226	(15)	
	PrecMax	4675	210	322	311	571	986	844	642	549	761	685	671	368	(16)	
	PrecMin	3006	1	28	121	158	161	164	93	106	154	215	66	73	(17)	
	PrecStd	418	55	72	56	96	215	144	152	123	186	115	140	84	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	34.7	35.2	34.7	33.7	32.2	32.6	32.1	32.2	32.8	32.7	32.7	(19)	
		MCWB	24.3	24.7	25.1	26.5	26.8	26.3	26.1	25.7	25.8	25.9	25.7	25.1	(20)	
	2%	DB	32.7	33.5	34.1	33.7	32.8	31.2	31.5	31.3	31.4	31.9	31.7	31.7	(21)	
		MCWB	23.9	24.7	25.4	26.7	26.8	26.1	25.7	25.6	25.5	25.8	25.7	24.9	(22)	
	5%	DB	31.9	32.6	33.2	32.8	32.0	30.6	30.7	30.5	30.8	31.0	30.9	30.9	(23)	
		MCWB	24.2	24.6	25.5	26.7	26.7	26.0	25.5	25.5	25.4	25.6	25.7	24.9	(24)	
	10%	DB	31.0	31.7	32.3	31.9	31.0	29.8	30.0	29.7	29.9	30.0	30.0	30.1	(25)	
		MCWB	24.3	24.7	25.5	26.5	26.6	25.9	25.3	25.4	25.4	25.5	25.6	25.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.4	26.7	27.1	28.0	27.9	27.3	26.8	26.7	26.7	26.9	26.7	26.7	(27)	
		MCDB	31.0	31.6	32.2	32.9	31.8	30.4	30.7	29.8	30.1	31.3	30.8	30.7	(28)	
	2%	WB	25.8	26.2	26.6	27.4	27.4	26.8	26.3	26.2	26.2	26.4	26.3	26.2	(29)	
		MCDB	30.2	31.0	31.7	32.1	31.2	29.7	29.9	29.5	29.5	30.4	30.2	30.0	(30)	
	5%	WB	25.4	25.7	26.2	27.0	27.1	26.5	26.0	25.9	26.0	26.0	26.0	25.7	(31)	
		MCDB	29.4	30.3	31.1	31.6	30.7	29.4	29.5	29.2	29.2	29.8	29.8	29.2	(32)	
	10%	WB	25.0	25.3	25.9	26.6	26.7	26.1	25.6	25.6	25.6	25.7	25.6	25.3	(33)	
		MCDB	28.8	29.6	30.5	30.7	29.9	29.0	28.9	28.6	28.6	28.9	28.9	28.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.6	9.7	8.6	6.6	5.3	5.6	5.7	6.2	7.0	7.4	7.6	(35)	
		MCDBR	10.5	10.9	10.6	9.7	8.2	6.4	6.8	7.1	7.6	8.4	8.5	9.2	(36)	
	5% WB	MCWBR	3.1	3.2	3.1	3.3	2.9	2.4	2.4	2.6	2.8	3.1	3.1	3.2	(37)	
		MCWBR	8.6	9.4	9.7	8.9	7.0	5.8	6.0	6.0	6.6	7.4	7.6	8.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.432	0.452	0.492	0.498	0.512	0.493	0.510	0.491	0.462	0.470	0.465	0.446	(40)	
		taud	2.346	2.276	2.179	2.190	2.143	2.212	2.137	2.208	2.308	2.277	2.289	2.327	(41)	
	Ebn at Noon		884	875	841	822	792	797	786	817	853	848	847	863	(42)	
		Edh at Noon	127	140	155	151	154	142	154	147	135	138	134	128	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.98	5.63	5.92	5.60	5.07	4.97	5.02	5.09	5.18	4.96	4.46	4.42	(44)	
	RadStd		0.38	0.33	0.34	0.30	0.38	0.30	0.22	0.25	0.36	0.30	0.30	0.39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		+0.17	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page